

SOFIANE SHAHIN

Embedded Systems & Avionics Engineering Student

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Engineering student in Embedded Systems (EFREI Paris, Intelligent Transportation Systems major), with a foundation in aeronautics from prep school (IPSA Paris). Holder of the BIA (aviation initiation certificate), currently training for a Private Pilot Licence (PPL). Seeking a 4-5 month internship, flexible availability from November 2, 2026 to April 2027, in avionics, embedded systems, or aerospace simulation, in France or Canada. Proven experience living and studying in Canada (academic exchange at Concordia University, Montreal).

SKILLS

Programming: Python • C • MATLAB • VHDL

Avionics & Embedded Systems: Multi-physics simulation, electronics, transmission systems, VHDL

Computer Vision & AI: OpenCV, template matching, HSV segmentation; working knowledge of Power BI / Azure DevOps

CAD & Design: CATIA • SolidWorks • PSpice

Technical Knowledge: Mechanics • Strength of Materials • Electronics • Thermodynamics • Aeronautics

Tools: Linux (Bash) • Git/GitHub • VS Code

Languages: French (native) • English (C1, TOEIC 860) • Spanish (intermediate)

Certifications: PPL (in progress) • BIA (French aviation initiation certificate) • Brown belt, Judo

EDUCATION

Master's in Engineering, Embedded Systems, Intelligent Transportation Systems major 2025 – 2028

Efrei Paris, France

- Coursework: modular avionics and onboard computers, multi-physics simulation (Thales partnership), future aircraft and flight management, autonomous navigation and environment perception (Valeo partnership)

International Academic Exchange in Computer Science Jan. – May 2026

Concordia University, Montreal, QC, Canada

- Coursework: operating systems architecture, computer architecture, networks, mathematics
- Hands-on experience living and studying in Canada, with practical understanding of the Quebec academic and professional environment

Preparatory Cycle in Aeronautics 2022 – 2025

IPSA Paris, France

- Coursework: mechanics, strength of materials, thermodynamics, aeronautics, electronics

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern Jul. – Aug. 2025

- Tracked strategic industrial modifications within the New Works department (DIPDE) at a nuclear power plant, a safety-critical environment transferable to aerospace certification standards
- Obtained safety clearance (PP58 protocol, 25 modules), authorizing interventions in controlled areas
- Project PNPE0339: tracked the integration of a water-spray cooling system for generator sets, in compliance with ASN (French Nuclear Safety Authority) requirements
- Participated in project steering meetings and drafted project tracking documentation

PROJECTS

Autonomous Robotic Navigation System Jun. – Jul. 2026

Efrei Paris

- Designed a multi-mission autonomous robot (Raspberry Pi, Python, OpenCV): infrared line following, camera tracking, obstacle avoidance
- Real-time sign detection via template matching and HSV segmentation for autonomous mission sequencing
- Integrated Picamera2 for real-time image processing and servo/motor control

MATLAB Simulations 2022 – 2025

IPSA Paris / Efrei Paris

- Developed simulations covering electronics and transmission systems

CAD & Mechanical Design Projects 2022 – 2025

IPSA Paris

- Designed mechanical components in CATIA for aerospace engineering applications